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A Experimental Details

A.1 Toy Setting: Random Projections Model

Setup. As a sanity check to empirically confirm our analytical predictions from Theorem 2, we consider a setting with multivariate Gaussian data (1). The feature covariance matrix Σ is constructed to have power-law eigenvalues $\lambda_j = C/j$, where C is such that $\text{tr } \Sigma = \lambda_1 + \dots + \lambda_d = 1$. The ground-truth labelling weights w_1^* of the real data distribution P_1 sampled from $N(0, (1/d)I_d)$, while the ground-truth weights w_2^* for the synthetic data distribution are sampled from $N(w_1^*, \Delta)$ with $\Delta = (c^2/d)\Sigma^{-1}$ for different values of c^2 ranging from $\{0, 0.1, 0.5, 1\}$ which controls for the quality of the synthetic data. We run a small experiment with label noise levels $\sigma_1 = \sigma_2 = 0.1$, input-dimension $d = 600$, number of real samples $n_1 = 300$, and synthetic samples $n_2 = 200$, for a total of $n = n_1 + n_2 = 500$ samples. We fit a random projection model $\hat{f}_{RP}$ according to (5) and for different values of the width parameter m (to control the size of the model), and report the results in Figures 5 and 4. The regularization parameter λ is set to a very small value (10^{-8}). We also consider a variation of this experiment with different values of the synthetic dataset size n_2 and report the results in Figure 1.

A.2 Two-layer neural networks

The two-layer neural networks are trained using stochastic gradient descent (SGD) with a batch size of 128 and a learning rate of 0.1. The models are trained for 400 epochs to fully converge. We employ a held-out validation set from the training set to select the best checkpoint to evaluate.

A.3 Language Modeling

The generation process for the BabiStories dataset is detailed in the GitHub repository of Zhang et al. (2024a). The dataset comprises a training set of 2,200,000 stories and a validation set of 22,000 stories, created by prompting the Mistral-8x7B model. Each prompt includes a description of the generation task, character names, specific words to be used, and the beginning of a story. The dataset stores the beginnings of these stories along with their generated continuations.

In our experiments, we trained a GPT-2-small model on this dataset to generate synthetic data. The model was trained using next-token prediction, utilizing the beginnings and continuations of stories to have good story generation quality. To maintain consistency with the original prompt distribution, we used all the prompts that were initially employed to generate BabiStories. During story generation, we applied a temperature setting of 1 with top-p decoding where $p = 1$. After generation, we filtered out stories of poor quality, such as those containing unwanted symbols, following the same procedure as in Zhang et al. (2024a). The filtered beginnings and synthetic continuations were then collected to form the synthetic dataset.

We used a GPT-2 model with an embedding dimension of $d = 768$, 12 attention heads, and a context length of 512 tokens, which typically encompasses one to three stories. During training, we applied a learning rate of 5×10^{-3} , a dropout rate of 0.05, L2 weight decay of 0.1, and a warm-up phase of 2,000 iterations.

B Some Omitted Theoretical Results and Comments

B.1 Classical Linear Model in Over-Parametrized Isotropic Setting

We now complement the analysis presented at the end of Section 3.1 with an analysis for the case $\phi \in (1, \infty)$. Plugging into Theorem 1 gives $\kappa \rightarrow \phi - 1$ and $u \rightarrow 1/(1 - \phi/\phi^2) = \phi/(\phi - 1)$ in the limit $\lambda \rightarrow 0^+$. We obtain the following corollary.

Corollary 4. *For $\phi \in (1, \infty)$, in the limit (4) and $\lambda \rightarrow 0^+$, it holds that $E_{test} \simeq \bar{E} + \zeta$, with*

$$\bar{E} = V + B, \quad B = r^2(1 - \frac{1}{\phi}), \quad V = \frac{\sigma^2}{\phi - 1}, \quad \zeta = \frac{p_2 c^2}{\phi^2} \left(p_2 \frac{\phi - p_2}{\phi - 1} + (\phi - p_2)^2 \right), \quad (20)$$

Moreover, for large $\phi \in (1, \infty)$, it holds that $E_{test}(\hat{f}_{CL}) - \bar{E} \simeq (1 - 2/\phi) p_2 c^2 + O(1/\phi^2)$.